Snubber Design Guidelines for a Half-Bridge Leg Based on the Infineon FZ2400R33HE4 IHM-B Module

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Abstract

This note collects the design guidelines used to size a snubber network for a half-bridge leg built with two 3300 V / 2400 A Trench/Fieldstop IGBT4 modules (Infineon FZ2400R33HE4, IHM-B package). It defines the voltage budget, quantifies the commutation-loop inductance, compares the practical snubber topologies, provides closed-form sizing equations, and works through a numerical example including loss estimation, component selection and test verification. A final section explains how to rescale the results to a 1700 V device of the same current class.

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1 Scope and device data

1.1 Note on the part number

The request referred to a *FZ2400R17* device, while the attached datasheet is the **FZ2400R33HE4**, i.e. a 3300 V class module. All numerical results in this note are derived from the attached datasheet (Revision 1.30, 2022-11-22). Section 11 gives the rescaling rules for a 1700 V module of the same current rating, for which the methodology is identical but the voltage budget and the resulting component values differ substantially.

1.2 Data relevant to snubber design

Only the parameters that enter the snubber calculation are reproduced here.

Table 1: FZ2400R33HE4 parameters relevant to snubber sizing

Parameter	Symbol	Value	Condition
Collector–emitter voltage	V_{CES}	3300 V	$T_{\mathrm{vj}} = -40$ to 150 °C
DC stability (cosmic ray)	$V_{CE(D)}$	2100 V	$T_{\mathrm{vj}} = 25$ °C, 100 FIT
Nominal DC collector current	$I_{C\,nom}$	2400 A	$T_C = 105$ °C
Repetitive peak current	I_{CRM}	4800 A	t_p limited by T_{vj}
Short-circuit current	I_{SC}	9600 A	$V_{CC} = 2400$ V, $t_p \le 10$ µs
Module stray inductance	L_{sCE}	6 nH	per switch
Test circuit stray inductance	L_σ	85 nH	datasheet switching test
Turn-off fall time	t_f	1.4 µs	150 °C, $R_{G,off} = 3.3\,\Omega$
Turn-off delay time	$t_{d,off}$	4.3 µs	150 °C
Turn-off $\mathrm{d}v/\mathrm{d}t$	$\dot{v}$	1500 V/µs	$T_{\mathrm{vj}} = 150$ °C
Turn-on $\mathrm{d}i/\mathrm{d}t$	i	7600 A/µs	$T_{\mathrm{vj}} = 150$ °C
Internal gate resistance	$R_{G,int}$	0.5 Ω	
Input capacitance	C_{ies}	280 nF	25 V, 1 MHz
Reverse transfer capacitance	C_{res}	8 nF	25 V, 1 MHz
Turn-off energy	E_{off}	4950 mJ	2400 A, 1800 V, 150 °C
Diode peak recovery current	I_{RM}	2880 A	150 °C, 7600 A/µs
Diode recovered charge	Q_r	2500 µC	150 °C
Diode minimum on-time	$t_{on,min}$	10 µs	

Two entries in Table 1 deserve emphasis because they dominate the design:

- The datasheet specifies the short-circuit rating as $V_{CE,\max} = V_{\mathrm{CES}} - L_{sCE}\,\mathrm{d}i/\mathrm{d}t$. The manufacturer therefore explicitly accounts for the inductive overshoot *inside* the module; every nanohenry added by the busbar reduces the usable blocking voltage in the same way.
- The DC stability limit $V_{CE(D)} = 2100$ V (100 FIT) caps the *steady-state* DC-link voltage. It is a cosmic-ray failure-rate limit, unrelated to the transient peak, and it must be checked independently.

2 The overvoltage mechanism

2.1 Commutation loop

At turn-off the collector current is interrupted in the loop formed by the DC-link capacitor, the busbar, the module terminals and the internal module leads. The total loop inductance

$$L_{loop} = L_{cap,ESL} + L_{busbar} + 2\,L_{sCE} \tag{1}$$

converts the current slope into a voltage transient across the device:

$$V_{\mathrm{pk}} = V_{\mathrm{DC}} + L_{loop}\left|\frac{\mathrm{d}i}{\mathrm{d}t}\right|_{\max} + \Delta V_{ring}. \tag{2}$$

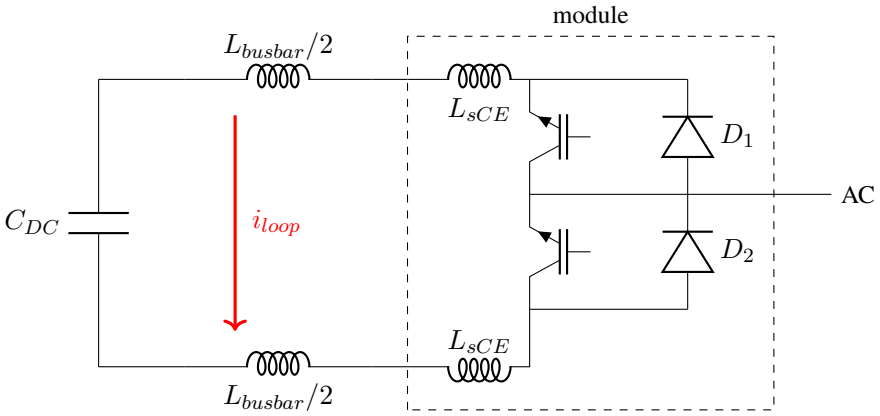


Figure 1: Commutation loop of the leg. The snubber must be inserted so that it shortens the loop seen by the switching device, i.e. as close as possible to the module DC terminals.

2.2 Which current slope applies

An IGBT does not chop the current instantaneously: the effective slope is set by the gate drive and by the device physics. Three operating cases have to be distinguished, and they lead to very different requirements.

Table 2: Current slope estimates for the three dimensioning cases

Case	Current	$\lvert\mathrm{d}i/\mathrm{d}t\rvert$	V_{DC}
Nominal turn-off	2400 A	$\approx 2.5\,\mathrm{kA/\mu s}$	1800 V
Overload turn-off	4800 A	$\approx 4\,\mathrm{kA/\mu s}$	1800 V
Short-circuit turn-off	9600 A	$\approx 8\,\mathrm{kA/\mu s}$	2400 V

For the nominal case the slope follows from the fall time, $\lvert\mathrm{d}i/\mathrm{d}t\rvert \approx k\,I_C/t_f$ with $k \approx 1.5\ldots2$ because the current waveform is not linear:

$$\left|\frac{\mathrm{d}i}{\mathrm{d}t}\right| \approx 1.5 \cdot \frac{2400\,\mathrm{A}}{1.4\,\mu\mathrm{s}} \approx 2.5\,\mathrm{kA/\mu s}. \tag{3}$$

The short-circuit case is far more aggressive: the desaturation protection turns off a current up to $I_{SC} = 9600\,\mathrm{A}$, and even with a soft turn-off gate drive the slope reaches several $\mathrm{kA/\mu s}$.

2.3 Diode recovery

The second overvoltage source is the reverse recovery of the complementary diode. With $Q_r = 2500\,\mu\mathrm{C}$ and $I_{RM} = 2880\,\mathrm{A}$ at $150\,^\circ\mathrm{C}$, the recovery current decay at the end of the recovery produces a transient across the diode. The emitter-controlled 4 diode is intentionally soft, but snappy behaviour can still appear at *low current* and *low junction temperature* — the worst case for the diode is therefore the opposite of the worst case for the IGBT. The datasheet minimum on-time $t_{on,min} = 10\,\mu\mathrm{s}$ must be respected, otherwise the recovery becomes uncontrolled.

3 Voltage budget

The design starts by allocating the available blocking voltage. Table 3 is the recommended allocation for a $3300\,\mathrm{V}$ module, following common industrial practice.

Table 3: Voltage budget for the $3300\,\mathrm{V}$ leg

Quantity	Value	Rationale
V_{CES}	$3300\,\mathrm{V}$	datasheet
Nominal DC link V_{DC}	$1800\,\mathrm{V}$	$\approx 0.55\,V_{\mathrm{CES}}$
Maximum DC link (regulation, braking)	$2000\,\mathrm{V}$	below $V_{CE(D)}$
Cosmic-ray limit $V_{CE(D)}$	$2100\,\mathrm{V}$	100 FIT, steady state
Design peak limit, normal operation	$2600\,\mathrm{V}$	$\approx 0.79\,V_{\mathrm{CES}}$
Design peak limit, fault turn-off	$3000\,\mathrm{V}$	$\approx 0.91\,V_{\mathrm{CES}}$, non-repetitive

From the peak limits the maximum admissible loop inductance follows directly by inverting (2):

$$L_{loop,\max} = \frac{V_{\mathrm{pk}}^{lim} - V_{\mathrm{DC}}}{|\mathrm{d}i/\mathrm{d}t|_{\max}}. \tag{4}$$

Table 4: Maximum admissible loop inductance per case

Case	$V_{\mathrm{pk}}^{lim} - V_{\mathrm{DC}}$	$\lvert\mathrm{d}i/\mathrm{d}t\rvert$	$L_{loop,\max}$
Nominal	$800\,\mathrm{V}$	$2.5\,\mathrm{kA/\mu s}$	$320\,\mathrm{nH}$
Overload	$800\,\mathrm{V}$	$4\,\mathrm{kA/\mu s}$	$200\,\mathrm{nH}$
Short circuit	$600\,\mathrm{V}$	$8\,\mathrm{kA/\mu s}$	$75\,\mathrm{nH}$

This is the central result of the analysis. In normal operation a well-designed laminated busbar (50 to $100\,\mathrm{nH}$) already satisfies the requirement with a comfortable margin, and *no snubber is strictly necessary*. The dimensioning case is the *short-circuit turn-off*, where the admissible loop inductance drops to $75\,\mathrm{nH}$ — a value that is very hard to reach with a real mechanical layout. The snubber (or an equivalent measure such as active clamping) is therefore justified primarily by the fault case and by the damping of the residual ringing, not by nominal operation.

Table 5: Comparison of snubber topologies for a high-power leg

Topology	Advantages	Drawbacks	Use
Pure C (decoupling)	Lossless, simple, two terminals, bolts directly onto the module	Undamped LC ringing with L_{loop}; no hard clamping	Standard, always present
Series RC across DC terminals	Damps the ringing, no diode required	Continuous loss, R must carry the full transient current	Common addition
RCD clamp per leg	Hard clamp of the peak, only active above the clamp level, protects during faults	Requires a fast $3300\,\mathrm{V}$ diode; bulky	Recommended here
RCD discharge-suppressing per switch	Also limits $\mathrm{d}v/\mathrm{d}t$	Capacitor becomes very large at $2400\,\mathrm{A}$; four components per switch	Not attractive
Undeland clamp	Single resistor for the whole leg, handles both IGBT and diode transients	More components, tuning of the discharge path	Alternative
Active clamping (gate feedback)	No power components, self-adapting, excellent for SCSOA	Increases E_{off}, requires a dedicated gate driver	Preferred modern alternative

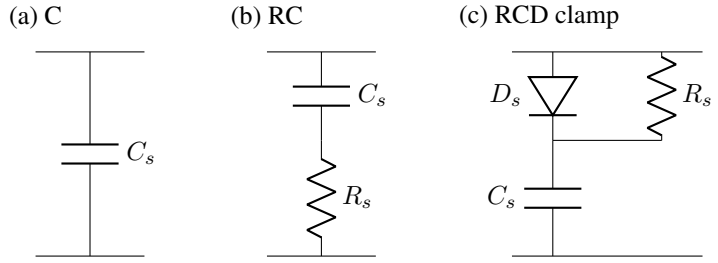


Figure 2: The three practical snubber configurations connected across the DC terminals of the leg. In (c) the resistor returns the clamped energy to the DC link.

4 Snubber topologies

5 Sizing equations

5.1 Pure capacitive snubber

With a capacitor C_s across the DC terminals, the loop becomes a resonant $L_{loop}C_s$ circuit with characteristic impedance

$$Z_0 = \sqrt{\frac{L_{loop}}{C_s}}, \qquad f_0 = \frac{1}{2\pi\sqrt{L_{loop}C_s}}. \tag{5}$$

If the device chopped the current instantaneously (worst case), the peak overshoot would be

$$\Delta V = I_{off}\, Z_0 = I_{off}\sqrt{\frac{L_{loop}}{C_s}} \quad \Longrightarrow \quad \boxed{C_s = \frac{L_{loop}\, I_{off}^2}{\Delta V^2}} \tag{6}$$

which is equivalent to the energy balance $\frac{1}{2}L_{loop}I_{off}^2 = \frac{1}{2}C_s\Delta V^2$. Equation (6) is conservative because a real IGBT dissipates most of the loop energy internally (this is precisely what E_{off} accounts for); it should be used as an upper bound.

Two secondary constraints must be checked:

$$\hat{i}_{C_s} = C_s \left.\frac{\mathrm{d}v}{\mathrm{d}t}\right|_{\max} \qquad \text{(peak capacitor current)} \tag{7}$$

$$f_0 \ll \frac{1}{2\pi\, t_f} \qquad \text{(resonance well above the switching transient)}$$

5.2 RC damping snubber

Adding a series resistor turns the loop into a damped second-order system with damping factor

$$\zeta = \frac{R_s}{2}\sqrt{\frac{C_s}{L_{loop}}} = \frac{R_s}{2Z_0}. \tag{8}$$

Critical damping requires $R_s = 2Z_0$; in practice $\zeta = 0.5 \ldots 1$, i.e.

$$\boxed{Z_0 \le R_s \le 2Z_0} \tag{9}$$

is a good compromise between ringing suppression and the additional voltage step $R_s\, I_{off}$ that the resistor itself injects. That step is the reason why R_s cannot be increased arbitrarily:

$$V_{\mathrm{pk}} \approx V_{\mathrm{DC}} + R_s I_{off} + \Delta V_C. \tag{10}$$

The dissipation depends on how much of the loop energy actually reaches the resistor. The pessimistic bound and the realistic estimate are

$$P_{R,\max} = \tfrac{1}{2}L_{loop}I_{off}^2\, n f_{sw} \qquad \text{(resistor absorbs the whole loop energy)} \tag{11}$$

$$P_{R,typ} = \tfrac{1}{2}C_s\Delta V_{ring}^2\, n f_{sw} \qquad \text{(resistor damps the residual ringing only)} \tag{12}$$

with n the number of damped commutations per switching period ($n = 2$ for a leg: IGBT turn-off and diode recovery).

5.3 RCD clamp

The clamp capacitor is pre-charged to V_{DC} through D_s and absorbs the loop energy only when the terminal voltage exceeds V_{DC}:

$$\tfrac{1}{2}L_{loop}I_{off}^2 = \tfrac{1}{2}C_s\left(V_{cl}^2 - V_{\mathrm{DC}}^2\right) \quad \Longrightarrow \quad \boxed{C_s = \frac{L_{loop}\,I_{off}^2}{V_{cl}^2 - V_{\mathrm{DC}}^2}} \tag{13}$$

The discharge resistor must return the stored charge to the DC link before the next commutation:

$$\tau = R_s C_s \le \frac{1}{5f_{sw}} \quad \Longrightarrow \quad \boxed{R_s \le \frac{1}{5f_{sw}C_s}} \tag{14}$$

and its dissipation, when the clamp is active, is

$$P_R = \tfrac{1}{2}L_{loop}I_{off}^2\, n f_{sw}. \tag{15}$$

The design intent is to set V_{cl} *above* the normal-operation peak so that the clamp does not conduct at all in steady state and (15) applies only during overload and fault events.

5.4 Discharge-suppressing RCD (for reference)

For the per-switch variant the capacitor follows from the requirement that the collector voltage does not rise faster than the current falls:

$$C_s \ge \frac{I_{off}\,t_f}{2\,\Delta V}. \tag{16}$$

At $I_{off} = 2400\,\mathrm{A}$, $t_f = 1.4\,\mu\mathrm{s}$ and $\Delta V = 400\,\mathrm{V}$ this gives $C_s \ge 4.2\,\mu\mathrm{F}$ *per switch*, which confirms that this topology is impractical at this current level.

6 Worked example

6.1 Assumptions

Table 6: Design assumptions

Quantity	Value
Nominal DC link V_{DC}	1800 V
Maximum DC link	2000 V
Switching frequency f_{sw}	500 Hz
Busbar inductance (laminated)	80 nH
Module contribution $2L_{sCE}$	12 nH
Capacitor ESL	8 nH
Total loop inductance L_{loop}	**100** nH

6.2 Unsnubbed peak voltage

Using (2) and Table 2:

$$\text{Nominal:} \quad V_{\mathrm{pk}} = 1800 + 100\times10^{-9}\cdot 2.5\times10^{9} = 1800 + 250 = 2050\,\mathrm{V} \tag{17}$$

$$\text{Overload:} \quad V_{\mathrm{pk}} = 1800 + 100\times10^{-9}\cdot 4\times10^{9} = 1800 + 400 = 2200\,\mathrm{V} \tag{18}$$

$$\text{Short circuit:} \quad V_{\mathrm{pk}} = 2400 + 100\times10^{-9}\cdot 8\times10^{9} = 2400 + 800 = 3200\,\mathrm{V} \tag{19}$$

Normal and overload operation stay well below the 2600 V design limit. The short-circuit turn-off exceeds the 3000 V fault limit and approaches V_{CES}: it requires either a clamp or an active-clamping gate driver.

6.3 Decoupling capacitor

Selecting a decoupling film capacitor $C_s = 1\,\mu\mathrm{F}$:

$$Z_0 = \sqrt{\frac{100\,\mathrm{nH}}{1\,\mu\mathrm{F}}} = 0.316\,\Omega \tag{20}$$

$$f_0 = \frac{1}{2\pi\sqrt{100 \times 10^{-9} \cdot 1 \times 10^{-6}}} = 503\,\mathrm{kHz} \tag{21}$$

The resonant frequency is two decades above the reciprocal of the fall time ($1/2\pi t_f \approx 114\,\mathrm{kHz}$), which is the correct separation. The peak current through the capacitor, from (7) with the datasheet $\mathrm{d}v/\mathrm{d}t = 1500\,\mathrm{V/\mu s}$, is

$$\hat{\imath}_{C_s} = 1\,\mu\mathrm{F} \cdot 1500\,\mathrm{V/\mu s} = 1500\,\mathrm{A}, \tag{22}$$

which is a hard constraint on the capacitor selection and the reason why a single high-$\mathrm{d}v/\mathrm{d}t$ film capacitor, or several in parallel, is required.

6.4 Damping resistor

$$R_s = Z_0 \ldots 2Z_0 = 0.32 \text{ to } 0.63\,\Omega \quad \Rightarrow \quad R_s = 0.5\,\Omega\ (\zeta = 0.79) \tag{23}$$

The voltage step introduced by the resistor at nominal current is $R_s I_{off} = 0.5 \times 2400 = 1200\,\mathrm{V}$, which is unacceptable if the resistor carries the full current. This is a classical trap: the *series* RC snubber must be sized with a much smaller R_s, or the resistor must be placed only in the damping path of an RCD structure. A practical compromise for a pure RC across the DC terminals is $R_s = 0.1$ to $0.2\,\Omega$ with $C_s = 2\,\mu\mathrm{F}$, accepting $\zeta \approx 0.35$ and a residual ringing of a few oscillations.

Dissipation from (12) with $\Delta V_{ring} = 250\,\mathrm{V}$, $n = 2$, $f_{sw} = 500\,\mathrm{Hz}$:

$$P_{R,typ} = \tfrac{1}{2} \cdot 2 \times 10^{-6} \cdot 250^2 \cdot 2 \cdot 500 = 62\,\mathrm{W}. \tag{24}$$

The pessimistic bound (11) gives $\frac{1}{2} \cdot 100 \times 10^{-9} \cdot 2400^2 \cdot 2 \cdot 500 = 288\,\mathrm{W}$, which sets the thermal rating of the resistor assembly.

6.5 RCD clamp for the fault case

Clamping level $V_{cl} = 2800\,\mathrm{V}$, $V_{\mathrm{DC}} = 2400\,\mathrm{V}$ (fault condition), $I_{off} = I_{SC} = 9600\,\mathrm{A}$, from (13):

$$C_s = \frac{100 \times 10^{-9} \cdot 9600^2}{2800^2 - 2400^2} = \frac{9.216}{2.08 \times 10^6} = 4.4\,\mu\mathrm{F} \tag{25}$$

Choose $C_s = 5\,\mu\mathrm{F}$ (film, 3000 V DC rating). Discharge resistor from (14):

$$R_s \le \frac{1}{5 \cdot 500 \cdot 5 \times 10^{-6}} = 80\,\Omega \quad \Rightarrow \quad R_s = 68\,\Omega \tag{26}$$

Since $V_{cl} = 2800\,\mathrm{V}$ is above the 2200 V overload peak, the clamp is idle in normal operation and the resistor only has to absorb the single-event energy

$$E_{ev} = \tfrac{1}{2} L_{loop} I_{SC}^2 = \tfrac{1}{2} \cdot 100 \times 10^{-9} \cdot 9600^2 = 4.6\,\mathrm{J}, \tag{27}$$

plus the energy already stored in C_s. A pulse-rated wirewound or thick-film resistor of a few tens of joules single-pulse capability is sufficient.

Table 7: Resulting snubber design for the leg

Function	Component	Value	Rating
Decoupling / damping	C_s	1 to 2 µF	2500 V DC, ESL < 20 nH, $\hat{\imath} > 2$ kA
Damping	R_s	0.1 to 0.2 Ω	non-inductive, 300 W
Clamp capacitor	C_{cl}	5 µF	3000 V DC film
Clamp diode	D_{cl}	–	4500 V, fast, soft recovery
Clamp resistor	R_{cl}	68 Ω	50 J single pulse

6.6 Summary of the design

7 Component selection

7.1 Capacitor

- Dry metallized polypropylene film, self-healing. Ceramic is unsuitable at this energy and voltage.
- DC voltage rating $\geq 1.3 \ldots 1.5\times$ the maximum applied DC voltage.
- $\mathrm{d}v/\mathrm{d}t$ rating: verify against (7). At $1500\,\mathrm{V/\mu s}$ a 1 µF capacitor sees 1500 A peak.
- ESL: the capacitor and its connection must contribute a small fraction of L_{loop}. Bolt-on terminals matching the module M8 studs, not leaded packages.
- RMS current: for the pure decoupling capacitor the RMS current is a few tens of amperes; verify against the ripple spectrum.

7.2 Resistor

- Non-inductive construction (thick film on ceramic, carbon composition, or bifilar wirewound). Series inductance < 50 nH, otherwise the resistor defeats its own purpose.
- Single-pulse energy rating from the fault case, average power rating from (11).
- Several resistors in parallel, symmetrically arranged, help both the thermal rating and the inductance.

7.3 Clamp diode

- $V_{RRM} \geq 4500$ V to keep margin over the clamp level.
- Fast and *soft* recovery: a snappy clamp diode generates its own overvoltage and can be worse than no clamp.
- Low forward recovery voltage V_{FR}: the diode must not overshoot during turn-on, otherwise the clamp is ineffective in the first microsecond, which is exactly when it is needed.

8 Layout guidelines

1. Laminated busbar with maximum overlap and minimum dielectric thickness compatible with the insulation coordination (Table 1 of the datasheet: creepage 32.2 mm, clearance 19.1 mm terminal-to-heatsink).
2. Snubber capacitor bolted directly to the module DC terminals, not to the DC-link capacitor. The purpose is to shorten the loop seen by the chip.
3. Symmetric arrangement if modules are paralleled: an asymmetry in L_{loop} translates into unequal turn-off overvoltage and unequal current sharing.
4. Keep the gate-drive return path separate from the power loop; the auxiliary emitter terminals of the IHM-B package exist for this reason.

5. Verify the loop inductance experimentally rather than trusting a calculation: from a double-pulse test, $L_{loop} = \Delta V/(\mathrm{d}i/\mathrm{d}t)$ measured at the terminals.

9 Alternatives to a passive snubber

Given the results of Table 4, two gate-side measures may replace or complement the snubber:

Active clamping

A chain of transient-voltage suppressors between collector and gate feeds current back into the gate when V_{CE} exceeds the clamp level, slowing the turn-off. It is compact, requires no bulky passives, and is very effective for the short-circuit case. The penalty is an increase of E_{off} and of the device junction temperature during the clamped event.

Two-level (soft) turn-off

The driver reduces the gate voltage to an intermediate level before switching off completely, or increases $R_{G,off}$ during a detected fault. With $R_{G,off}$ increased from $3.3\,\Omega$ to $10\,\Omega$ the fault $\mathrm{d}i/\mathrm{d}t$ can typically be halved, which relaxes (4) by the same factor. The datasheet switching-loss and switching-time curves $E = f(R_G)$ and $t = f(R_G)$ quantify the trade-off.

Increasing $R_{G,off}$ is the cheapest lever available and should be evaluated before adding copper and components: it costs switching energy but no volume.

10 Verification plan

1. **Loop inductance measurement.** Double-pulse test at low voltage; extract L_{loop} from $\Delta V/(\mathrm{d}i/\mathrm{d}t)$ and compare with the FEM/analytical busbar estimate.
2. **Nominal double-pulse test.** $V_{\mathrm{DC}} = 2000\,\mathrm{V}$, $I_C = 2400\,\mathrm{A}$, $T_{\mathrm{vj}} = 150\,^{\circ}\mathrm{C}$. Verify $V_{\mathrm{pk}} \leq 2600\,\mathrm{V}$ and check the waveform against the datasheet $t_{d,off}$, t_f and E_{off}.
3. **Overload test.** $I_C = I_{CRM} = 4800\,\mathrm{A}$; verify the trajectory stays inside the RBSOA diagram of the datasheet.
4. **Diode recovery test.** Low current (10 to 20 % of I_{nom}) and low temperature ($25\,^{\circ}\mathrm{C}$) to expose snappy recovery; verify $t_{on,min} \geq 10\,\mu\mathrm{s}$ is respected by the modulator.
5. **Short-circuit tests.** Type I and Type II at $V_{\mathrm{DC}} = 2400\,\mathrm{V}$, $t_p \leq 10\,\mu\mathrm{s}$; verify V_{pk} against $V_{\mathrm{CES}} - L_{sCE}\,\mathrm{d}i/\mathrm{d}t$ and check the clamp energy.
6. **Thermal check of the snubber.** Measure the resistor temperature in continuous operation at maximum modulation index and compare with the estimate from (12).

11 Rescaling to a 1700 V module

If the intended device is a $1700\,\mathrm{V}$ / $2400\,\mathrm{A}$ module, the methodology is unchanged but the numbers shift as follows.

Table 8: Voltage budget comparison

Quantity	3300 V class	1700 V class
V_{CES}	3300 V	1700 V
Typical DC link	1800 V	900 V
Design peak, normal operation	2600 V	1350 V
Available headroom $V_{\mathrm{pk}} - V_{\mathrm{DC}}$	800 V	450 V
Typical t_f	$\approx 1.4\,\mu\mathrm{s}$	$\approx 0.4\,\mu\mathrm{s}$

Two effects act in the same, unfavourable direction:

- the available headroom is roughly halved;

- the $1700\,\mathrm{V}$ device is intrinsically faster, so $\mathrm{d}i/\mathrm{d}t$ at the same current is two to three times higher.

From (4) the admissible loop inductance therefore drops by roughly a factor of five, into the 20 to $40\,\mathrm{nH}$ range for nominal operation. For a $1700\,\mathrm{V}$ leg at $2400\,\mathrm{A}$ the busbar design becomes the dominant engineering effort, and the decoupling capacitor is mandatory rather than optional. The clamp capacitor from (13) scales as $1/(V_{cl}^2 - V_{\mathrm{DC}}^2)$ and grows substantially: a recalculation with the actual datasheet values of the chosen part is required.

12 Conclusions

1. For the FZ2400R33HE4 at $V_{\mathrm{DC}} = 1800\,\mathrm{V}$ with a $100\,\mathrm{nH}$ commutation loop, the nominal turn-off overvoltage is about $250\,\mathrm{V}$, giving a peak of $2050\,\mathrm{V}$ against a $2600\,\mathrm{V}$ design limit. A snubber is *not* required for overvoltage limitation in normal operation.
2. A decoupling film capacitor of 1 to $2\,\mu\mathrm{F}$ directly on the module terminals is nevertheless mandatory, to shorten the commutation loop and to keep the resonance well separated from the switching transient.
3. The dimensioning case is the short-circuit turn-off, which would require $L_{loop} \leq 75\,\mathrm{nH}$. This is addressed by an RCD clamp set at $2800\,\mathrm{V}$, by active clamping, or by a soft turn-off gate drive — the last being the cheapest option.
4. The series resistance of an RC snubber is constrained by the voltage step $R_s I_{off}$; at $2400\,\mathrm{A}$ this forces $R_s \leq 0.2\,\Omega$ and accepts partial damping.
5. All results must be confirmed by a double-pulse test at maximum voltage, maximum current and $150\,^{\circ}\mathrm{C}$, plus a short-circuit test at $2400\,\mathrm{V}$.

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